Multimodal Medical Assistant

Capstone Project Report — Project ID: HPPCS[04]

Abstract

This project implements a Multimodal Medical Assistant that pairs a medical image with its corresponding text prescription/patient details and generates a grounded, multi-turn clinical conversation as JSON output. The pipeline uses two LLMs, each specialized for a distinct modality: a vision-language model to interpret the medical image, and a text-generation LLM to combine those findings with the prescription/patient text and produce the final conversation. Both models run locally via Ollama, requiring no API key. The system is evaluated on five image–text input pairs, producing one JSON conversation file per pair (five output files total). This report covers the problem, objectives, model choices, methodology, and evaluation approach.

1. Introduction

Clinicians routinely interpret medical images (X-rays, MRIs, scans) alongside a patient's written prescription or clinical details, manually connecting visual findings to textual context before forming an assessment. This project builds a Multimodal Medical Assistant that automates this connection: given a medical image paired with its corresponding text prescription/patient details, the system generates a structured clinician-style conversation that reasons jointly over both inputs. The assistant is implemented as an individual capstone project (Project ID: HPPCS04) and evaluated on five image–text input pairs.

2. Problem Statement

Medical image interpretation and textual clinical context are typically processed by separate tools, requiring manual cross-referencing by a clinician. There is a need for a single pipeline that accepts a medical image and its associated text prescription/patient details as a pair, reasons across both modalities, and produces an evidence-grounded, conversational output that a clinician could review — without requiring the two data types to be interpreted independently.

3. Objectives

- Accept five medical images, each paired with a corresponding text prescription/patient-details file, as project input.
- Interpret each medical image using a vision-capable model and interpret/structure the paired text using a language model.
- Fuse the two modalities so the generated output reflects both the visual finding and the textual patient context together, rather than either in isolation.

- Generate a multi-turn, clinician-style conversation for each of the 5 input pairs, grounded in the image and the prescription/patient details.
- Output the conversation for each input pair as a separate JSON file (5 output files total), suitable for downstream review or evaluation.

4. Models Used and Justification

The project uses two LLMs, both served locally through Ollama, so no API key or paid access is required, as required by the assignment guidelines:

- Vision-language model — "llava" (via Ollama): used to interpret the medical image and produce a structured description of visible findings (e.g., anomalies, affected region, notable features). Chosen because it is a free, open-source vision-language model that runs fully offline and is queried through the same simple local API as the text model.
- Text-generation LLM — "llama3" (via Ollama): used to read the patient's text prescription/details, combine them with the image model's structured findings, and generate the final multi-turn conversation, formatted as JSON. Chosen because it is free, runs fully offline, and is strong at fluent, instruction-following text generation.

Splitting the pipeline across two specialized models — one for visual grounding and one for language generation — keeps each model within its area of strength, avoids relying on a single general-purpose model to perform both fine-grained image interpretation and fluent multi-turn dialogue generation, and keeps the entire pipeline free to run with no external API dependency.

5. Methodology / System Design

For each of the 5 input pairs, the pipeline proceeds in three stages. First, the medical image is passed to the vision-language model, which returns a structured textual summary of relevant visual findings. Second, this summary is combined with the corresponding text prescription/patient details and passed to the text-generation LLM, which is prompted to produce a realistic clinician–patient (or clinician-to-clinician) conversation that references both the image finding and the prescription context. Third, the resulting conversation is parsed and written to a JSON output file with a consistent schema (e.g., a list of speaker-turn objects), one file per input pair, giving five output files in total.

The pipeline is implemented in Python with a single entry point, main.py, which accepts the input and output directories as parameters and writes the five JSON conversation files to the output location. No API key is required, since both models run locally through Ollama. This is in line with the submission guidelines (single directory, no sub-directories, all files accessible via "./").

6. Results and Analysis

For each of the five test cases, the pipeline is expected to produce one JSON conversation file that (a) references a specific visual finding from the image, (b) incorporates details from the corresponding text prescription, and (c) reads as a coherent multi-turn exchange. Evaluation focuses on whether the generated conversation is grounded in both modalities (rather than ignoring one), whether the JSON output is well-formed and consistently structured across all five files, and whether the conversation content is clinically plausible given the inputs.

7. Conclusion and Future Work

This project implements a Multimodal Medical Assistant that pairs a medical image with its corresponding text prescription/patient details and generates a grounded, structured conversation as JSON output, using two complementary LLMs — one specialized for medical image interpretation and one for language generation. Future extensions could include validating generated conversations against clinician review, extending the pipeline to additional imaging modalities (e.g., MRI slices, pathology slides), and adding automated consistency checks between the image finding and the generated dialogue.

8. References

Relevant open-source model documentation was consulted for the vision-language and text-generation components used in this project, including Ollama documentation and model cards for the "llava" and "llama3" models.

The Llama 3 Herd of Models (Llama 3 & 3.1 Technical Report)

- Citation: Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, S., Letman, A., Mathur, A., Schelten, A., Yang, A., et al. (2024). The Llama 3 Herd of Models. arXiv preprint arXiv:2407.21783.
- Paper: https://arxiv.org/abs/2407.21783
- Meta Documentation: https://llama.meta.com/docs/model-cards-and-prompt-formats/

LLaVA-NeXT (Stronger Open Multimodal Reasoning)

- Citation: Liu, H., Li, C., Li, Y., Li, B., Zhang, Y., Shen, S., & Lee, Y. J. (2024). *LLaVA-NeXT: Improved reasoning, OCR, and world knowledge*.
- Technical Release: https://llava-vl.github.io/blog/2024-01-30-llava-next/

Acknowledgement

I would like to express my sincere gratitude to all the professors of IIT Kharagpur and the TCS iON moderators for their guidance, support, and valuable feedback throughout the course of this project.